BIAN-STYLE SERVICE LANDSCAPE · PRODUCT-NEUTRAL

AI Platform Capability Model

Every agent platform is the same 127 services. Fourteen of them no platform ships.

Three tiers: **8 capability areas** are the columns, **25 capabilities** the labeled groups, and **127 services** the tiles; each service is exactly one functional pattern applied to one asset, written `Pattern · Asset`, for example `Agree · Folder`.

8	25	127	14
AREAS	CAPABILITIES	SERVICES	OPEN

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BA1 Session & Turn

5 capabilities · 20 services

Ability to hold a conversation with the agent as a durable, resumable record, run each turn inside it under the user's control, and recover it after interruption, across every surface the platform offers.

Session Store
Session runtime team · BA1.01

- **Session Catalog** — Register · Session — A searchable list of every conversation, with identity and timestamps.
- **Transcript Log** — Maintain · Transcript — The append-only record of what was said and done in a session.
- **Workspace Binding** — Register · Workspace — The directory a session acts in, recorded so it can be resumed in place.
- **Attachment Store** — Maintain · Attachment — Files a session attached or produced, kept beside the transcript.
- **External Reference** — Register · Reference — Issues, pull requests, and other typed links attached to a turn.

Turn Loop
Agent runtime team · BA1.02

- **Turn Execution** — Fulfill · Turn — One cycle of prompt, model call, tool calls, and response.
- **Tool Call Dispatch** — Process · Tool Call — Routing a model's tool request to the right tool and returning the result.

Turn Control
Agent runtime team · BA1.03

- **Turn Interrupt** — Operate · Turn — Stopping a running turn cleanly, from the keyboard or over a protocol.
- **Message Steering** — Process · Inbound Message — Injecting guidance into a turn that is already running.
- **Human Input Request** — Agree · Question — The agent pausing to ask the user a question or for a decision.
- **Run & Spend Limit** — Allocate · Budget — A cap on turns, time, tool calls, or money for one run.

Continuity
Session runtime team · BA1.04

- **Resume Set** — Track · Session — Which sessions are open or pinned, so they come back on restart.
- **Session Search** — Analyze · Transcript — Full-text search across past transcripts.
- **Session Fork** — Provision · Child Session — Branching a conversation into a child session.
- **Progress Checkpoint** — Process · Checkpoint — A written summary of work done and next steps at a point in time.
- **Session Share Link** — Fulfill · Share Link — OPEN — A link that lets another person open a session.

Recovery
Session runtime team · BA1.05

- **File Rewind** — Maintain · File Snapshot — Restoring files to how they were before an agent edit.
- **Context Compaction** — Process · Context — Shrinking the context window by summarizing older turns.
- **Archive & Erasure** — Administer · Session — Archiving or deleting sessions and their traces.
- **Transcript Export** — Fulfill · Export — OPEN — Writing a session out in a portable format.

SCOPE

Covers the session record, the turn loop, in-flight control, continuity, and recovery. Excludes what the agent is told before a turn (Memory) and what it is allowed to do during one (Authority & Configuration).

BA2 Authority & Configuration

4 capabilities · 19 services

Ability to bound what the agent may do, as which identity, under which policy, and to configure that at user, project, and organization scope, with every grant recorded.

Admission
Trust and policy team · BA2.01

- **Workspace Trust** — Agree · Folder — A recorded decision that the agent may act in a folder.
- **Managed Policy** — Administer · Policy — Organization or device policy that overrides user settings.
- **Content Exclusion** — Administer · Exclusion Rule — Rules that keep named files or paths out of the agent's reach.
- **Untrusted Content Guard** — Assess · Content — OPEN — Marking pasted, fetched, or downloaded content so it is treated as data, not instructions.

Identity
Identity and accounts team · BA2.02

- **Account & Credential** — Register · Account — Signing in to the vendor or holding the key for an endpoint.
- **Data Use Consent** — Agree · Consent — The user's choice about telemetry and training on their data.
- **Agent Credential** — Register · Agent Identity — OPEN — A scoped identity of the agent's own, distinct from the user's login.

Execution Control
Trust and policy team · BA2.03

- **Sandbox Policy** — Administer · Sandbox — The filesystem and process fence a tool runs inside.
- **Network Egress** — Administer · Network Rule — Which hosts the agent and its tools may reach.
- **Approval Mode** — Administer · Approval — The stance on when to ask before acting: always, never, or by risk.
- **Remembered Grant** — Register · Grant — An allow or deny decision kept so the same action is not asked again.
- **Permission Stream** — Track · Permission — The live record of each permission asked and answered.
- **Secret Redaction** — Process · Secret — OPEN — Scrubbing credentials from transcripts, logs, and model input.
- **Secret Injection** — Process · Credential — OPEN — Supplying a credential to one tool call and clearing it afterward.
- **Content Safety** — Assess · Output — OPEN — A filter on model output for harmful content.
- **Audit Export** — Fulfill · Audit Record — OPEN — An auditor-ready record of what the agent did and who allowed it.

Configuration
Platform configuration team · BA2.04

- **User Settings** — Administer · Setting — The user's own configuration file for the platform.
- **Project Settings** — Administer · Project Setting — Configuration scoped to one repository or directory.
- **Feature Flag** — Administer · Flag — Switches that turn experimental behavior on or off.

SCOPE

Covers admission, identity, execution control, and configuration. Excludes the tools themselves (Extension & Tools) and the model provider's own limits (Models).

BA3 Extension & Tools

3 capabilities · 20 services

Ability to act on the world through native tools and to extend the agent with installable packages and configured behavior, without changing the platform's own code.

Native Tools
Tooling team · BA3.01

- **Shell Execution** — Operate · Terminal — Running commands in a terminal on the user's behalf.
- **File Edit** — Operate · File — Reading, creating, and changing files in the workspace.
- **Code Search** — Analyze · Codebase — Fast text and symbol search over the codebase.
- **Web Retrieval** — Process · Web Page — Searching the web and fetching pages as context.
- **Browser & Computer Use** — Operate · Desktop — Driving a browser or desktop as a tool.
- **Code Interpreter** — Operate · Interpreter — A REPL the model can run code in.
- **Language Service** — Analyze · Symbol — Symbols, diagnostics, and navigation from a language server.
- **Codebase Index** — Maintain · Index — OPEN — A maintained semantic index of the workspace for retrieval.

Packages
Extension ecosystem team · BA3.02

- **Skill Pack** — Register · Skill — A folder of instructions and scripts the agent can load on demand.
- **Plugin Marketplace** — Administer · Plugin — A catalog to discover and install extensions from.
- **MCP Server** — Administer · Server — A configured server that exposes tools and resources over the Model Context Protocol.
- **Skill Mining** — Analyze · Trajectory — Proposing new skills from observed tool trajectories.
- **Plugin Verification** — Assess · Plugin — OPEN — Checking a package's signature or hash before it is installed.
- **MCP Sampling** — Fulfill · Sampling — OPEN — Letting a tool server ask the client to run the model.

Behavior
Extension ecosystem team · BA3.03

- **Slash Command** — Operate · Command — A named command the user types to trigger a prompt or action.
- **Workflow Script** — Fulfill · Workflow — A scripted multi-step procedure the agent executes.
- **Persona Definition** — Design · Persona — A description of how the agent should behave and sound.
- **Role Assignment** — Allocate · Role — Giving an agent a defined responsibility, such as reviewer or implementer.
- **Lifecycle Hook** — Process · Event — User code that runs on events such as tool start or file change.
- **Output Style** — Design · Style — A preference for language, tone, or format of responses.

SCOPE

Covers native tools, packages, and behavior. Excludes the permission to use a tool (Authority & Configuration) and the context assembled for it (Memory).

BA4 Memory

3 capabilities · 14 services

Ability to brief the agent before each request and to retain, retrieve, and forget knowledge across sessions, with a known precedence and a bounded context window.

Briefing
Context engineering team · BA4.01

- **Prompt Assembly** — Process · Prompt — Composing instruction files, memory, and dynamic context into one request.
- **Instruction File** — Design · Instruction — A markdown file the agent reads before starting.
- **Instruction Precedence** — Design · Context Order — The order in which global, project, and local instructions win.
- **System Prompt Override** — Administer · System Prompt — Replacing or extending the system prompt from the command line.
- **Dynamic Context** — Process · Context Item — Repository and branch snippets added to the prompt automatically.
- **Context Budget** — Allocate · Context Window — OPEN — Choosing what enters the context window against a size limit.

Retention
Memory team · BA4.02

- **Memory Index** — Maintain · Memory — The top-level file that lists what the agent remembers.
- **Topic Note** — Register · Note — One remembered fact or observation, kept as its own entry.
- **Knowledge Base** — Maintain · Knowledge — A structured store of what the agent has learned about a project.
- **Team Memory** — Agree · Memory — Notes shared by everyone who works in a repository.
- **User Profile** — Register · Profile — What the agent knows about the person it works with.

Pipeline
Memory team · BA4.03

- **Memory Maintenance** — Process · Memory — Consolidating, deduplicating, and refining stored memory.
- **Memory Retrieval** — Analyze · Memory — Finding the relevant memory for the current task.
- **Memory Erasure** — Administer · Memory — Forgetting on request, with tombstones so it stays forgotten.

SCOPE

Covers briefing, retention, and the memory pipeline. Excludes the transcript of a session (Session & Turn), which is a record rather than a memory.

BA5 Change & Review

2 capabilities · 8 services

Ability to produce changes to a codebase that are staged, attributed, isolated, reviewed, and shipped through the repository's own workflow.

Change Set
Source control integration team · BA5.01

- **Change Staging** — Maintain · Changeset — Collecting an agent's edits into a reviewable set.
- **Edit Attribution** — Track · File Edit — Recording which files a session touched and with which tool.
- **Commit Authoring** — Fulfill · Commit — Writing commit messages and attributing commits to the agent.
- **Worktree Isolation** — Administer · Worktree — Giving a session its own git worktree and branch.
- **Repository State** — Monitor · Repository — Tracking the branch, remote, and git status a session works against.

Review
Code review team · BA5.02

- **Review Annotation** — Assess · Annotation — Comments pinned to lines of a change.
- **Code Review** — Assess · Change — An agent judging a change for correctness and quality.
- **Pull Request Flow** — Fulfill · Pull Request — Opening a pull request and tracking its status.

SCOPE

Covers the change set and its review. Excludes the file-edit tool itself (Extension & Tools) and the plan that led to the change (Orchestration).

BA6 Orchestration

4 capabilities · 15 services

Ability to plan, delegate, queue, trigger, and reach work above the level of a single turn, across agents, time, and machines.

Planning
Orchestration team · BA6.01

- **Execution Plan** — Design · Plan — A plan written and agreed before the agent acts.
- **Goal & Budget** — Track · Goal — The objective of a run and the tokens and time allotted to it.

Delegation
Orchestration team · BA6.02

- **Subagent Delegation** — Allocate · Agent — Handing a sub-task to a child agent with its own context.
- **Foreign Agent Hosting** — Provision · Agent SDK — Running another vendor's agent inside this platform's host.
- **Competing Runs** — Assess · Trajectory — Running several attempts at once and picking the best.
- **Peer Message** — Process · Message — Agents sending messages to each other.

Work Queue
Orchestration team · BA6.03

- **Issue Intake** — Register · Issue — Accepting an issue or ticket as the start of a run.
- **Follow-up Queue** — Maintain · Queue — Messages queued to run after the current turn.
- **Scheduled Job** — Operate · Schedule — A run that starts on a timer.
- **Event Trigger** — Operate · Trigger — A run that starts when something happens.
- **Work Board** — Manage · Work Item — A board of work items the agent and user share.

Reach
Remote and cloud team · BA6.04

- **Remote Control** — Operate · Endpoint — Driving a local agent from another device or network.
- **Cloud Run** — Provision · Cloud Session — The vendor running the agent on its own infrastructure.
- **Shared Room** — Operate · Room — A hosted space where several people and agents work together.
- **Fleet Console** — Monitor · Fleet — One view over many running agents.

SCOPE

Covers planning, delegation, the work queue, and reach. Excludes the single-turn loop (Session & Turn) and the surfaces the user meets (Surfaces & Operations).

BA7 Models

2 capabilities · 11 services

Ability to supply model capacity from catalogs, providers, and local runtimes, choose and route it per request, and account for its cost against a plan.

Model Supply
Model gateway team · BA7.01

- **Model Catalog** — Register · Model — The list of models available to this platform.
- **Model Selection** — Allocate · Model — Which model a session uses, and switching it mid-session.
- **Provider Routing** — Allocate · Provider — Sending requests to a chosen provider or endpoint.
- **Local Runtime** — Operate · Runtime — Serving models from the user's own machine.
- **Custom Model Build** — Provision · Model — Building a named model variant from a base and a recipe.
- **Bring-your-own Key** — Administer · Credential — Using the user's own provider key instead of the vendor's billing.
- **Retry & Rate Limit** — Process · Request — Backing off and retrying when the provider throttles or fails.

Cost
Model gateway team · BA7.02

- **Generation Control** — Administer · Generation Setting — Reasoning effort and sampling parameters for a request.
- **Usage Account** — Analyze · Usage — Tokens, calls, and cost recorded per session and model.
- **Prompt Cache** — Process · Cache — Reusing cached prompt prefixes to cut cost and latency.
- **Plan & Entitlement** — Agree · Subscription — The subscription or quota that governs what the user may consume.

SCOPE

Covers model supply and cost. Excludes the agent's reasoning behavior that uses the model (Session & Turn) and the user's sign-in (Authority & Configuration).

BA8 Surfaces & Operations

2 capabilities · 20 services

Ability to meet people and programs wherever they are, from terminal to protocol to chat, and to keep the installation observable, current, and healthy.

Surfaces
Client surfaces team · BA8.01

- **Command Line** — Operate · CLI — The interactive terminal client.
- **Headless Run** — Operate · Run — A non-interactive run that takes a prompt and emits structured output.
- **Agent Protocol Server** — Provision · Protocol Endpoint — Serving the agent to other programs over a documented protocol.
- **Embeddable SDK** — Provision · SDK — A library for building the agent into another product.
- **IDE Attach** — Operate · IDE — Connecting to an editor so the agent sees and edits what the user has open.
- **Messaging Channel** — Operate · Channel — Reaching the agent through a chat platform.
- **Voice** — Process · Audio — Speaking to the agent, or the agent speaking back.
- **Multimodal Input** — Process · Image — Giving the agent images and screenshots as input.
- **Document Authoring** — Fulfill · Document — Producing documents, spreadsheets, slides, and PDFs.
- **Media Processing** — Process · Media — Transforming audio, video, and images.
- **Media Generation** — Fulfill · Media — OPEN — Generating images or video from a prompt.
- **Canvas Document** — Design · Canvas — A rich, editable document the agent and user build together.
- **Desktop & Push Notification** — Operate · Notification — Telling the user when the agent needs them or has finished.
- **Web & Mobile Surface** — Operate · Web App — OPEN — Using the agent from a browser or phone.

Care
Platform operations team · BA8.02

- **Runtime Log** — Monitor · Log — Process logs for debugging the platform itself.
- **Telemetry & Crash** — Monitor · Telemetry — Usage telemetry and crash reports sent to the vendor.
- **Version & Update** — Maintain · Install — Knowing the installed version and moving to the next one.
- **Gateway Liveness** — Monitor · Gateway — Heartbeats that show a channel or connection is alive.
- **Health Check** — Assess · Installation — A self-check that reports what is misconfigured.
- **Feedback Capture** — Register · Feedback — Sending a bug report or feedback from inside the product.

SCOPE

Covers surfaces and care. Excludes the session a surface opens (Session & Turn) and remote or cloud execution (Orchestration).

PREVALENCE ACROSS THE 9 REVIEWED PLATFORMS

- **Wide** 16 — Most clients store it
- **Mixed** 36 — Several clients, different shapes
- **Rare** 59 — One or two clients
- **Host** 2 — The model runtime
- **Open** 14 — In the product class, but no reviewed platform ships it

EVIDENCE BASE · 9 PLATFORMS REVIEWED 2026-10-18

Cx Codex OpenAI · Cl Claude Code Anthropic · Cp GitHub Copilot GitHub · Ge Gemini CLI Google · Ag Antigravity Google · Gk Grok xAI · He Hermes Nous Research · Qw Qwen Alibaba · Ol Ollama Ollama · model host

INTERACTIVE MODEL

claude.ai/artifact/N1aTBgXoraEoz2PqjavbS5

AI Platform Capability Model v3
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